

The total possible points are 80:

P1 (15). (a): 5. (b) 7. (c) 3; the problem is fully specified for obtaining the ratio between two mass feed rates.

P2 (10). 10; deriving the right DOF worth 4 points and getting the right volumetric feed is 6 points.

P3 (25). (a) 12 points: each feed worth 4 points. (b) 6 points. (c) 7 points; as long as you can realize that the problem is over-specified, 4 points are offered.

P4 (15). Getting the right for CO₂ consumption worth 6 points. Having a reasonable logic for estimating CO₂ consumption worth 6 points. A comment on the difference between calculation and estimation worth 3 points. Generally, an error of 30% can be considered reasonable.

P5 (15). Obtaining the right DOF worth 6 points. Realizing the problem is under-specified worth 3 points. Getting the right flow recommendations worth 6 points; it needs show the goal is to minimize total cost.